distribution. This includes sanitation, pasteurization, holding conditions and microbiological standards. The U.S public health secrine publication "Milk ordinance and code" shows the following chemical, bacteriological and temperature standards for grade A milk and milk products.

### 5.5.5 Methylene Blue dye Reduction Test (MBRT)

Methylene blue dye reduction test commonly known as MBRT test is used as a quick method to access the microbiological quality of raw and pasteurized milk. This test is based on the fact that the blue colour of the dye solution added to the milk get decolorized when the oxygen present in the milk get exhausted due to microbial activity. The sooner the de colorization, more inferior is the bacteriological quality of milk assumed to be MBRT test may be utilized for grading of milk which may be useful for the milk processor to take a decision on further processing of milk.

**Procedure**

The test has to be done under sterile conditions. Take 10ml milk sample in sterile MBRT test tube. Add 1 ml Methylene Blue dye solution (dye concentration 0.005%). Stopper the tubes with sterilized rubber stopper and carefully place them in a test tube stand dipped in a serological water bath maintained at 37 1°C, records this time as the beginning of the incubation period.

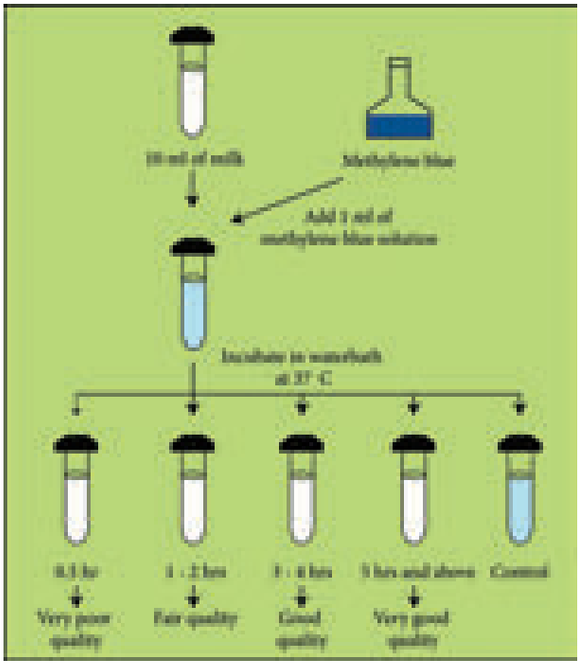



**Figure 5.4:** Methylene Blue dye Reduction Test

Decolourization is considered complete when only a faint blue ring (about 5mm) persists at the top (Figure 5.4).

Recording of Results – During incubation, observe colour changes as follows:

a. If any sample is decolourized on incubation for 30 minutes, record the reduction time as MBRT – 30 minutes.

b. Record such readings as, reduction times in whole hours. For example, if the colour disappears between 0.5 and 1.5 hour readings, record the result as MBRT – 1 hour, similarly, if between 1.5 and 2.5 hours as MBRT-2 hour and so on.

**Table 5.5:** Microbiological Quality of Milk

| S.No | Grade | Methylene blue reductase test in hrs | Total plate count/ml |
|---|---|---|---|
| 1. | Very good | 5 and above | Not exceeding 0.2 million |
| 2. | Good | 3–4 | Between 0.2 to1.0 million |
| 3. | Fair | 1–2 | Between 1–5 million |
| 4. | Very poor | 0.5 | over 5 million |



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